

```

#include <stdio.h>
#include <stdlib.h>

int global = 100;      // Initialized Global variable -> Data Segment
static int s = 50;     // Static variable -> Data Segment

int main()
{
    int stack = 10;     // Local variable -> Stack Segment
    int *heap = (int*)malloc(sizeof(int)); // Dynamic Allocation -> Heap Segment

    // Output memory addresses of various process segments
    printf("Code Segment : %p\n", (void*)main);
    printf("Global Segment: %p\n", (void*)&global);
    printf("Static Segment: %p\n", (void*)&s);
    printf("Heap Segment : %p\n", (void*)heap);
    printf("Stack Segment : %p\n", (void*)&stack);

    printf("\nProgram paused. Press ENTER to exit...\n");
    getchar(); // Keeps process alive to inspect /proc/PID/maps

    free(heap); // Free heap memory
    return 0;
}

```

^h Get Help
^x Exit

^o WriteOut
^u Justify

^R Read File
^W Where is

^Y Prev Pg
^Y Next Pg

^K Cut Text
^u UnCut Text

^C Cur
^t To S

Parent=100

medikondarakesh@Medikondas-MacBook-Air ~ % nano memory_layout.c

medikondarakesh@Medikondas-MacBook-Air ~ % gcc memory_layout.c -o memory_layout

medikondarakesh@Medikondas-MacBook-Air ~ % ./memory_layout

Code Segment : 0x100ac84f8

Global Segment: 0x100ad0000

Static Segment: 0x100ad0004

Heap Segment : 0x100f9d170

Stack Segment : 0x16f337158

Program paused. Press ENTER to exit...

medikondarakesh@Medikondas-MacBook-Air ~ %